

Problem 4.65

A horizontal constant-diameter pipe with a build-up of debris is shown in Fig. P4.65. Air modeled as an ideal gas enters at 320 K, 900 kPa, with a velocity of 30 m/s and exits at 305 K. Assuming steady state and neglecting stray heat transfer, determine for the air exiting the pipe

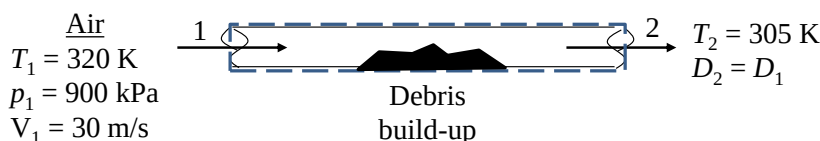
(a) the velocity, in m/s.

(b) the pressure, in kPa.

KNOWN: Air flows at steady state through a constant-area pipe section with a build-up of debris. Data are known at the inlet and exit.

FIND: For the air exiting the pipe section, determine the velocity and the pressure.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL:

1. The control volume shown on the schematic is at steady state.
2. Air is modeled as an ideal gas.
3. For the control volume, $\dot{W}_{cv} = 0$, $\dot{Q}_{cv} = 0$, and $\Delta pe = 0$.

ANALYSIS:

(a) The energy rate balance for the one-inlet, one-exit control volume at steady state

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}[(h_1 - h_2) + \frac{1}{2}(V_1^2 - V_2^2) + g(z_1 - z_2)]$$

simplifies to

$$0 = (h_1 - h_2) + \frac{1}{2}(V_1^2 - V_2^2)$$

Solving for exit velocity gives

$$V_2 = \sqrt{V_1^2 + 2(h_1 - h_2)}$$

From Table A-22, $h_1 = 320.29 \text{ kJ/kg}$ and $h_2 = 305.22 \text{ kJ/kg}$. Solving for exit velocity

$$V_2 = \sqrt{\left(30 \frac{\text{m}}{\text{s}}\right)^2 + 2\left(320.29 \frac{\text{kJ}}{\text{kg}} - 305.22 \frac{\text{kJ}}{\text{kg}}\right)\left|\frac{1000 \text{ N} \cdot \text{m}}{1 \text{ kJ}}\right|\left|\frac{1 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}}{1 \text{ N}}\right|}$$

$$V_2 = 176.2 \text{ m/s}$$

Problem 4.65 (Continued)

(b) From the mass rate balance and ideal gas equation of state

$$\dot{m}_2 = \dot{m}_1 = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} = \frac{\cancel{A_1} V_1 p_1}{\cancel{R} T_1} = \frac{\cancel{A_2} V_2 p_2}{\cancel{R} T_2}$$

Since the inlet and exit diameters are the same, the areas cancel as well as the gas constant.
Solving for exit pressure yields

$$p_2 = p_1 \left(\frac{V_1}{V_2} \right) \left(\frac{T_2}{T_1} \right)$$

$$p_2 = (900 \text{ kPa}) \left(\frac{30 \text{ m/s}}{176.2 \text{ m/s}} \right) \left(\frac{305 \text{ K}}{320 \text{ K}} \right)$$

$$p_2 = 146.1 \text{ kPa} \quad \longleftarrow$$

Problem 4.70

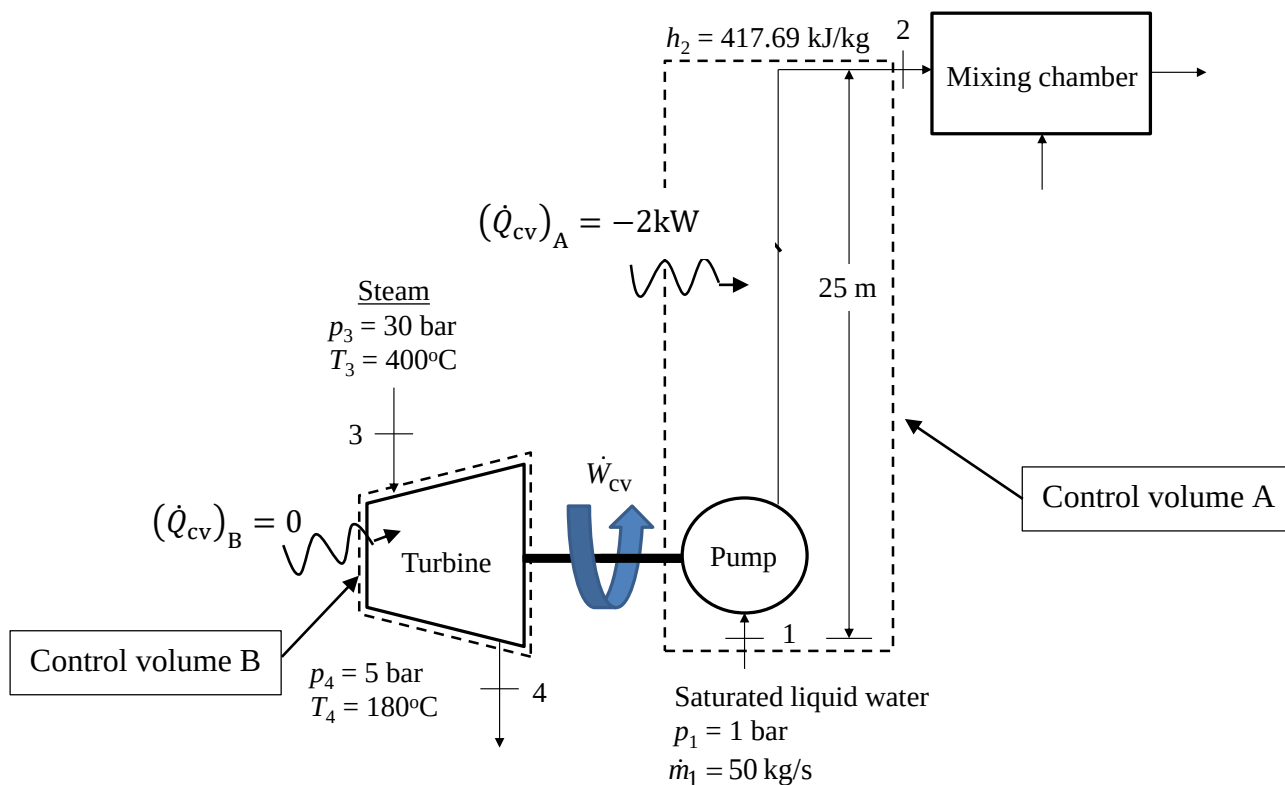
Figure P4.70 shows a turbine-driven pump that provides water to a mixing chamber located 25 m higher than the pump. Steady-state operating data for the turbine and pump are labeled on the figure. Heat transfer from the water to its surroundings occurs at a rate of 2 kW. For the turbine, heat transfer with the surroundings and potential energy effects are negligible. Kinetic energy effects at all numbered states can be ignored. Determine

- (a) The power required by the pump, in kW, to supply water to the inlet of the mixing chamber.
- (b) The mass flow rate of steam, in kg/s, that flows through the turbine.

KNOWN: A steam turbine drives a pump through which water flows to a mixing chamber located 25 m higher than the pump. Data are provided at the inlets and exits of the turbine and the pump.

FIND: (a) The power required by the pump, in kW, to supply water to the inlet of the mixing chamber and (b) the mass flow rate of steam, in kg/s, that flows through the turbine.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL:

1. All components operate at steady state.
2. Define control volume A to encompass the pump and the line to the mixing chamber.
3. Define control volume B to encompass the turbine.
4. For both control volumes A and B, kinetic energy effects can be ignored.
5. No stray heat transfer occurs between control volume B and its surroundings.
6. For control volume B, potential energy effects can be ignored.

Problem 4.70

ANALYSIS:

(a) At steady state, the mass balance for Control volume A reduces to: $\dot{m}_1 = \dot{m}_2 \equiv \dot{m}_{\text{water}}$
The energy rate balance at steady state for control volume A

$$0 = \dot{Q}_{\text{cv}} - \dot{W}_{\text{cv}} + \dot{m}_{\text{water}} [(h_1 - h_2) + \cancel{1/2 (V_1^2 - V_2^2)} + g(z_1 - z_2)]$$

simplifies to

$$\dot{W}_{\text{cv}} = \dot{Q}_{\text{cv}} + \dot{m}_{\text{water}} [(h_1 - h_2) + g(z_1 - z_2)]$$

The specific enthalpy at state 1 is obtained from Table A-3: $h_1 = h_{f1} = 417.46 \text{ kJ/kg}$. Since the elevation at state 2 is higher than the elevation at state 1, $(z_1 - z_2) = -25 \text{ m}$. Substituting values and solving yield

$$\dot{W}_{\text{cv}} = -2 \text{ kW} + \left(50 \frac{\text{kg}}{\text{s}} \right) \left[\left(417.46 \frac{\text{kJ}}{\text{kg}} - 417.69 \frac{\text{kJ}}{\text{kg}} \right) + \left(9.81 \frac{\text{m}}{\text{s}^2} \right) (-25 \text{ m}) \right] \left| \frac{1 \text{ N}}{1 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}} \right| \left| \frac{1 \text{ kJ}}{1000 \text{ N} \cdot \text{m}} \right| \left| \frac{1 \text{ kW}}{1 \frac{\text{kJ}}{\text{s}}} \right|$$

$$\dot{W}_{\text{cv}} = -25.8 \text{ kW} \quad \leftarrow$$

Since the value for power is negative, work is into the pump as expected.

(b) At steady state, the mass balance for Control volume B reduces to: $\dot{m}_3 = \dot{m}_4 \equiv \dot{m}_{\text{steam}}$ The energy rate balance for control volume B

$$0 = \cancel{\dot{Q}_{\text{cv}}} - \dot{W}_{\text{cv}} + \dot{m}_{\text{steam}} [(h_3 - h_4) + \cancel{1/2 (V_3^2 - V_4^2)} + g(\cancel{z_3} - z_4)]$$

simplifies to

$$0 = -\dot{W}_{\text{cv}} + \dot{m}_{\text{steam}} (h_3 - h_4)$$

since $\dot{m}_3 = \dot{m}_4$. Solving for mass flow rate of steam yields

$$\dot{m}_{\text{steam}} = \frac{\dot{W}_{\text{cv}}}{h_3 - h_4}$$

Since both states 3 and 4 are superheated vapor, their specific enthalpies are obtained from Table A-4: $h_3 = 3230.9 \text{ kJ/kg}$ and $h_4 = 2812.0 \text{ kJ/kg}$. Since the turbine produces the power required by the pump, $\dot{W}_{\text{cv}} (\text{turbine}) = -\dot{W}_{\text{cv}} (\text{pump}) = -(-25.8 \text{ kW}) = 25.8 \text{ kW}$. Substituting values and solving yield

$$\dot{m}_{\text{steam}} = \frac{25.8 \text{ kW}}{3230.9 \frac{\text{kJ}}{\text{kg}} - 2812.0 \frac{\text{kJ}}{\text{kg}}} \left| \frac{1 \frac{\text{kJ}}{\text{s}}}{1 \text{ kW}} \right|$$

$$\dot{m}_{\text{steam}} = 0.0616 \text{ kg/s} \quad \leftarrow$$

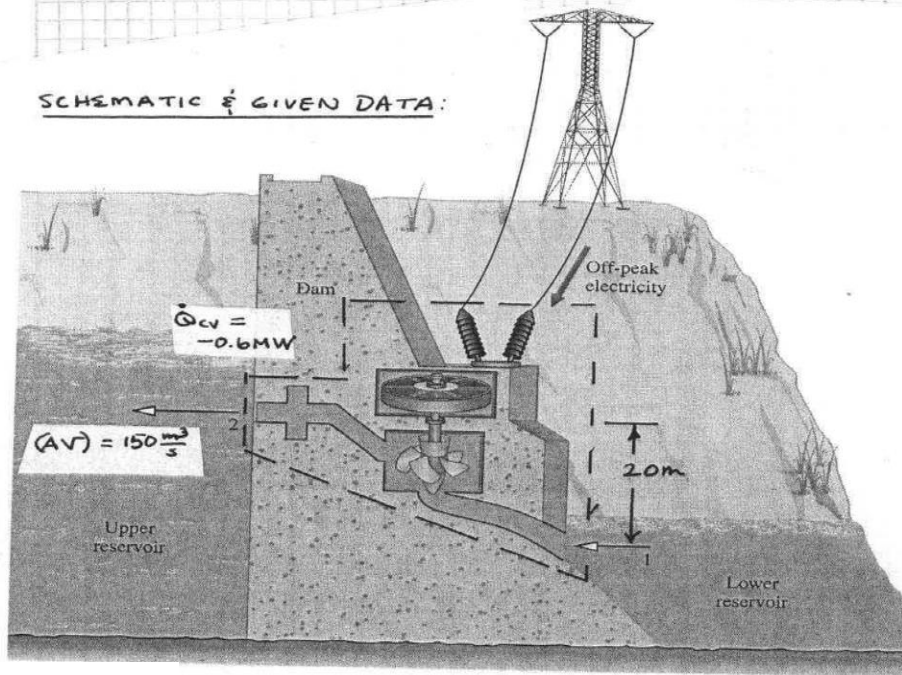
Problem 4.72

Figure 4.72 shows a *pumped-hydro* energy storage system delivering water at steady state from a lower reservoir to an upper reservoir using *off-peak* electricity. Water is delivered to the upper reservoir at a volumetric flow rate of $150 \text{ m}^3/\text{s}$ with an increase in elevation of 20 m . There is no significant change in temperature, pressure, or kinetic energy from inlet to exit. Heat transfer from the pump to its surroundings occurs at a rate of 0.5 MW and $g = 9.81 \text{ m/s}^2$. Determine the pump power required, in MW . Assuming the same volumetric flow rate when the system generates *on-peak* electricity using this water, will the power be greater, less, or the same as the pump power? Explain.

KNOWN: Steady-state data are provided for a pumped-hydro system delivery water from a lower reservoir to a higher reservoir using off-peak electricity.

FIND: Determine the pump power required, in MW . Discuss.

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL:

1. The control volume shown in the schematic is at steady state.
2. For the control volume there is no significant change in temperature, pressure, or kinetic energy from inlet to exit.
3. $g = 9.81 \text{ m/s}^2$
4. Water is modeled as incompressible with $\rho = 1000 \text{ kg/m}^3$ (Table A-19).

ANALYSIS: The mass rate balance reduces to give, $\dot{m}_1 = \dot{m}_2 (= \dot{m})$.

The energy rate balance reduces as follows:

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[\underbrace{(h_1 - h_2)}_{=0} + \underbrace{\frac{(V_1^2 - V_2^2)}{2}}_{=0} + g(z_1 - z_2) \right]$$

$= 0 \text{ (Eq. 3.20b with } T_2 \sim T_1 \text{ and } P_2 \sim P_1)$

$$\Rightarrow \dot{W}_{cv} = \dot{Q}_{cv} + \dot{m} g(z_1 - z_2) \quad , \text{ where } \dot{m} = \rho(AV) .$$

$$= -0.6 \text{ MW} + \left[\left(\frac{1000 \text{ kg}}{\text{m}^3} \right) \left(150 \frac{\text{m}^3}{\text{s}} \right) \left(9.81 \frac{\text{m}}{\text{s}^2} \right) (-20 \text{ m}) \right] \left| \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right| \left| \frac{1 \text{ MW}}{10^6 \text{ N} \cdot \text{m/s}} \right|$$

① $= -30 \text{ MW} \quad . \text{ Thus, the power required} = 30 \text{ MW}$

The heat transfer in this instance can be traced to frictional effects, including electrical resistance. Such effects also will be present as the control volume generates on-peak electricity using stored water. Accordingly, for the same volumetric flow rate, the rate power is developed will be less than 30 MW .

1. In the absence of heat transfer from the system the power required is 29.4 MW .

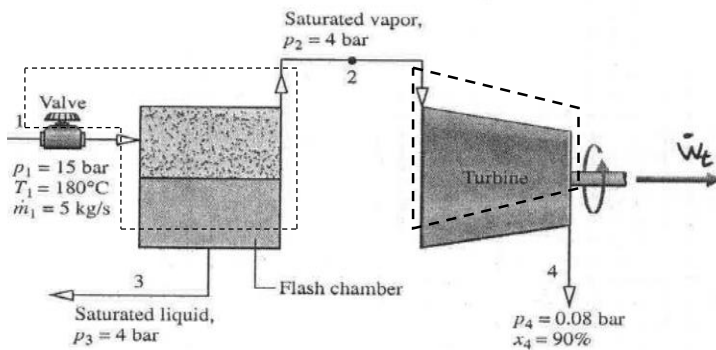
Problem 4.75

As shown in fig. P4.75, hot industrial waste water at 15 bar, 180°C with a mass flow rate of 5 kg/s enters a flash chamber via a valve. Saturated vapor and saturated liquid streams, each at 4 bar, exit the flash chamber. The saturated vapor enters the turbine and expands to 0.08 bar, $x = 90\%$. Stray heat transfer and kinetic and potential energy effects are negligible. For operation at steady state, determine the power, in hp, developed by the turbine.

KNOWN: Steady-state data are provided for three components in series: valve, flash chamber, and turbine.

FIND: Determine the power developed by the turbine, in hp.

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL:

1. Control volumes enclosing the turbine and the valve plus flash chamber are considered.
2. The control volumes are at steady state.
3. For each control volume, $\dot{Q}_{cv}$ is negligible and kinetic and potential energy effects are ignored.

ANALYSIS: For a control volume enclosing the turbine, mass and energy rate balance combine to give, $\dot{W}_t = \dot{m}_2 (h_2 - h_4)$. (1)

Since $\dot{m}_2$ is unknown, a control volume enclosing the valve and flash chamber is considered. The mass rate balance reduces to give $\dot{m}_1 = \dot{m}_2 + \dot{m}_3 \Rightarrow \dot{m}_3 = \dot{m}_1 - \dot{m}_2$. An energy rate balance reduces with assumptions listed as follows: $0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 h_1 - \dot{m}_2 h_2 - \dot{m}_3 h_3$. Then, with $\dot{m}_3 = \dot{m}_1 - \dot{m}_2$, we get

$$0 = \dot{m}_1 h_1 - \dot{m}_2 h_2 - (\dot{m}_1 - \dot{m}_2) h_3$$

Solving,

$$\dot{m}_2 = \frac{\dot{m}_1 [h_1 - h_3]}{[h_2 - h_3]}$$

With $h_1 \approx h_f(180^\circ\text{C})$ from Table A-2, and $h_2 = h_g$ and $h_3 = h_f$ at 4 bar from Table A-3, we get

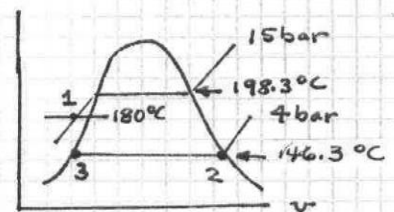
$$\dot{m}_2 = \frac{5 \text{ kg/s} [763.22 - 604.74] \text{ kJ/kg}}{[2738.6 - 604.74] \text{ kJ/kg}} = 0.371 \text{ kg/s}$$

In Eq. (1), h_4 is also needed: $h_4 = h_f + x(h_g - h_f)$ where h_f and h_g are at 0.08 bar

$$\therefore h_4 = 173.88 + 0.9 [2577 - 173.88] = 2336.69 \text{ kJ/kg}$$

Substituting values into Eq. (1),

$$\begin{aligned} \dot{W}_t &= (0.371 \text{ kg/s}) [2738.6 - 2336.69] \frac{\text{kJ}}{\text{kg}} = 149.1 \frac{\text{kJ}}{\text{s}} \left| \frac{1 \text{ kW}}{1 \text{ kJ/s}} \right| = 149.1 \text{ kW} \\ &= 149.1 \text{ kW} \left| \frac{1 \text{ hp}}{0.7457 \text{ kW}} \right| = 199.9 \text{ hp} \end{aligned}$$



Problem 4.87

An open cooking pot containing 0.5 liter of water at 20°C, 1 bar sits on a stove burner. Once the burner is turned on the water is gradually heated at a rate of 0.85 kW while pressure remains constant. After a period of time, the water starts boiling and continues to do so until all of the water has evaporated. Determine

(a) the time required for the onset of evaporation, in s.

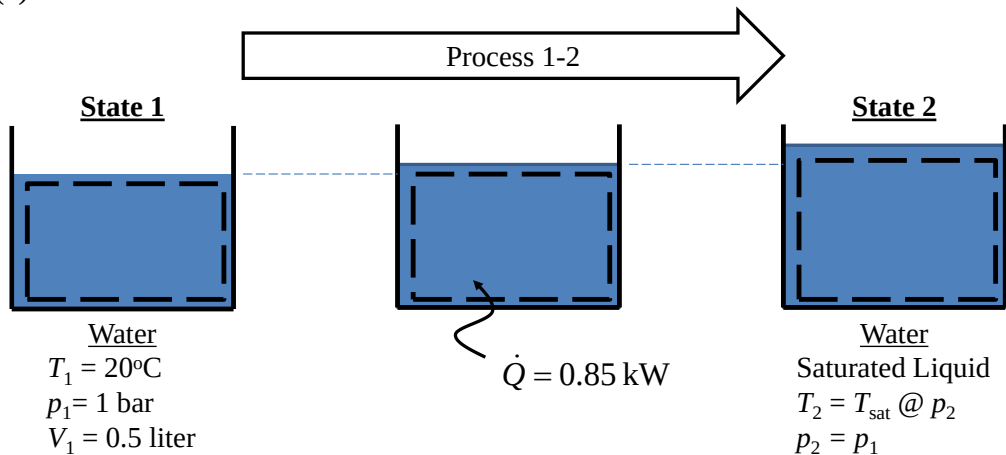
(b) the time required for all of the water to evaporate, in s, once evaporation starts.

KNOWN: Water is heated at constant pressure in an open pot on a burner. The constant rate of heat transfer is specified.

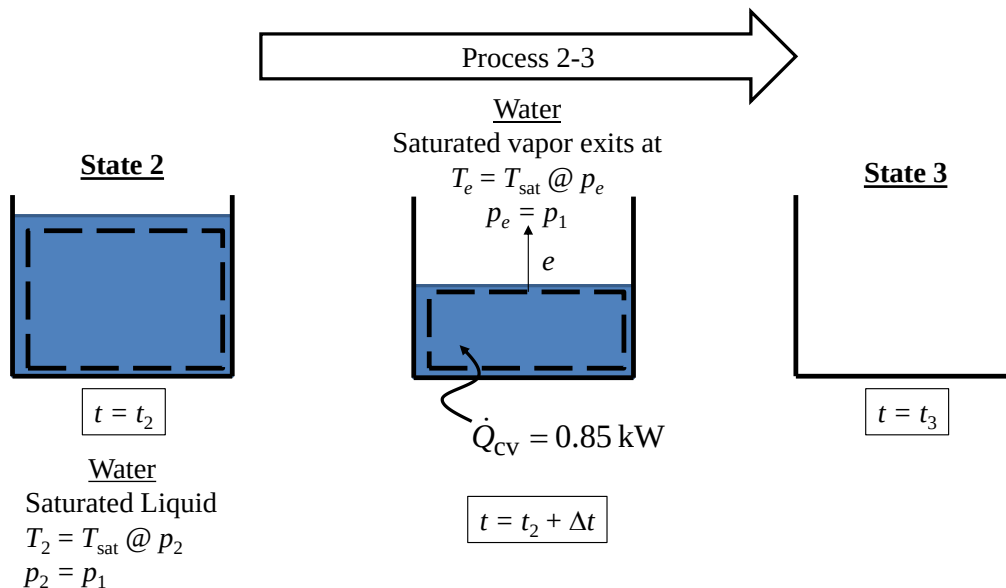
FIND: Determine the time required for the onset of evaporation, and the time required for all of the water to evaporate, once evaporation starts.

SCHEMATIC AND GIVEN DATA:

Part (a)

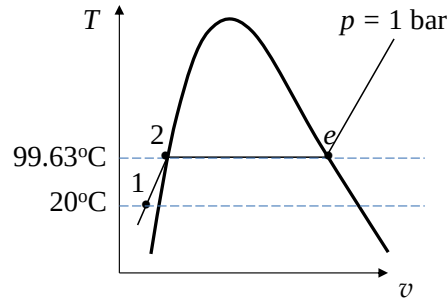


Part (b)



Problem 4.87 (Continued – Page 2)

The temperature-specific volume diagram below indicates the states associated with the overall process:



ENGINEERING MODEL:

1. For part (a) the liquid water inside the pot defines the system during process 1-2 as shown by the dashed line on the accompanying figure. For part (b) the control volume is defined by the boundary of the liquid water inside the pot during process 2-3 as shown by the dashed line on the accompanying figure.
2. For the system and control volume, kinetic and potential effects can be ignored.
3. As shown in the figure of part (b), saturated vapor exits the control volume at 1 bar.
4. Water is assumed to remain at 1 bar throughout the entire process.
5. Work associated with system boundary movement (part a) and work associated with control volume boundary movement (part b) are not ignored.

ANALYSIS:

(a) During process 1-2, heat transfer results in a temperature increase from 20°C to 99.63°C (the saturation temperature that corresponds to 1 bar). Since evaporation has not yet started, there is no mass flow of water into or from the system, and the water can be analyzed as a closed system. Since the mass is constant during process 1-2 and with assumption 2 in the engineering model, the energy rate balance for process 1-2 reduces to

$$\textcircled{1} \quad \Delta U = Q - W$$

or

$$m(u_2 - u_1) = \dot{Q}\Delta t - W$$

Solving for Δt gives

$$\Delta t = \frac{m(u_2 - u_1) + W}{\dot{Q}}$$

The mass of water can be determined from the volume of the liquid water and its corresponding specific volume at state 1. At $p_1 = 1$ bar and $T_1 = 20^\circ\text{C}$, water is compressed liquid. From Table A-2, $v_1 \approx v_{f1} = 0.0010018 \text{ m}^3/\text{kg}$

$$m = \frac{V_1}{v_1} = \frac{(0.5 \text{ L})}{0.0010018 \frac{\text{m}^3}{\text{kg}}} \left| \frac{10^{-3} \text{ m}^3}{1 \text{ L}} \right| = 0.50 \text{ kg}$$

The specific internal energy at state 1 is determined from Table A-2, $u_1 \approx u_{f1} = 83.95 \text{ kJ/kg}$. The water is saturated liquid at state 2. From Table A-3 at $p_2 = 1$ bar, $u_2 = u_{f2} = 417.36 \text{ kJ/kg}$.

As the liquid water is heated, its volume increases slightly. Thus, *work is done by the system at the top surface* during process 1-2

Problem 4.87 (Continued – Page 3)

$$W = \int_1^2 p dV = mp(v_2 - v_1)$$

From Table A-3, $v_2 = v_{f2} = 0.0010432 \text{ m}^3/\text{kg}$. Solving for work

$$W = (0.50 \text{ kg})(1 \text{ bar})(0.0010432 \frac{\text{m}^3}{\text{kg}} - 0.0010018 \frac{\text{m}^3}{\text{kg}}) \left| \frac{100 \text{ kPa}}{1 \text{ bar}} \right| \left| \frac{1000 \frac{\text{N}}{\text{m}^2}}{1 \text{ kPa}} \right| \left| \frac{1 \text{ kJ}}{1000 \text{ N} \cdot \text{m}} \right|$$

$$W = 0.00207 \text{ kJ}$$

Substituting values and solving for Δt give

$$\Delta t = \frac{(0.50 \text{ kg})(417.36 \frac{\text{kJ}}{\text{kg}} - 83.95 \frac{\text{kJ}}{\text{kg}}) + 0.00207 \text{ kJ}}{0.85 \text{ kW}} \left| \frac{1 \text{ kW}}{1 \frac{\text{kJ}}{\text{s}}} \right| = 196.1 \text{ s (3.27 min)} \leftarrow$$

(b) During process 2-3, heat transfer results in phase change from liquid to vapor at 99.63°C (the saturation temperature that corresponds to 1 bar). Mass flow from the control volume is the saturated vapor resulting from evaporation of the liquid. The mass rate balance takes the form

$$\frac{dm_{\text{cv}}}{dt} = -\dot{m}_e$$

The energy rate balance for process 2-3 reduces to

$$\frac{dU_{\text{cv}}}{dt} = \dot{Q}_{\text{cv}} - \dot{W}_{\text{cv}} - \dot{m}_e h_e$$

Combining mass and energy rate balances results in

$$\frac{dU_{\text{cv}}}{dt} = \dot{Q}_{\text{cv}} - \dot{W}_{\text{cv}} + \frac{dm_{\text{cv}}}{dt} h_e$$

By assumption 3 of the engineering model, the specific enthalpy at the exit is constant. Accordingly, integration gives

$$\Delta U_{\text{cv}} = \dot{Q}_{\text{cv}} \Delta t - \dot{W}_{\text{cv}} \Delta t + \Delta m_{\text{cv}} h_e$$

$$m_3 u_3 - m_2 u_2 = \dot{Q}_{\text{cv}} \Delta t - \dot{W}_{\text{cv}} \Delta t + (m_3 - m_2) h_e$$

Mass at state 2 is the system mass during process 1-2: $m_2 = 0.50 \text{ kg}$. Since no mass remains in the control volume at state 3, $m_3 = 0$. The equation reduces to

$$\textcircled{2} \quad -m_2 u_2 = \dot{Q}_{\text{cv}} \Delta t - \dot{W}_{\text{cv}} \Delta t - m_2 h_e$$

Solving for time associated with evaporation during process 2-3 yields

$$\Delta t = \frac{m_2 (h_e - u_2) + \dot{W}_{\text{cv}} \Delta t}{\dot{Q}_{\text{cv}}}$$

Problem 4.87 (Continued – Page 4)

The specific enthalpy for saturated vapor at the exit is obtained from Table A-3 at $p_e = 1$ bar:
 $h_e = h_{ge} = 2675.5$ kJ/kg.

The liquid volume decreases as vapor exits the control volume. Thus, *work is done on the control volume at the moving portion of the boundary* during process 2-3

$$W_{cv} = \int_2^3 p dV = p(V_3 - V_2) = -pV_2 = -m_2 p_2 v_2$$

Substituting values and solving for work give

$$W_{cv} = -(0.50 \text{ kg})(1 \text{ bar})(0.0010432 \frac{\text{m}^3}{\text{kg}}) \left| \frac{100 \text{ kPa}}{1 \text{ bar}} \right| \left| \frac{1000 \frac{\text{N}}{\text{m}^2}}{1 \text{ kPa}} \right| \left| \frac{1 \text{ kJ}}{1000 \text{ N} \cdot \text{m}} \right|$$
$$W_{cv} = -0.0522 \text{ kJ}$$

Solving for time associated with evaporation during process 2-3 yields

$$\Delta t = \frac{(0.50 \text{ kg})(2675.5 \frac{\text{kJ}}{\text{kg}} - 417.36 \frac{\text{kJ}}{\text{kg}}) + (-0.0522 \text{ kJ})}{0.85 \text{ kW}} \left| \frac{1 \text{ kW}}{1 \frac{\text{kJ}}{\text{s}}} \right| = 1328.3 \text{ s (22.1 min)} \leftarrow$$

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- ① For process 1-2 a combination of the energy balance and the work expression gives an expression for energy represented in terms of specific enthalpy only.

$$m(u_2 - u_1) = \dot{Q}\Delta t - W = \dot{Q}\Delta t - mp(v_2 - v_1)$$

$$m[(u_2 + pv_2) - (u_1 + pv_1)] = \dot{Q}\Delta t$$

$$m(h_2 - h_1) = \dot{Q}\Delta t$$

- ② For process 2-3 a combination of the energy balance and the work expression gives an expression for energy represented in terms of specific enthalpy only.

$$-m_2 u_2 = \dot{Q}_{cv}\Delta t - W_{cv} - m_2 h_e = \dot{Q}_{cv}\Delta t - (-m_2 p_2 v_2) - m_2 h_e$$

$$-m_2(u_2 + p_2 v_2) = \dot{Q}_{cv}\Delta t - m_2 h_e$$

$$-m_2 h_2 = \dot{Q}_{cv}\Delta t - m_2 h_e$$

$$-m_2(h_2 - h_e) = \dot{Q}_{cv}\Delta t$$